Tutorial 1

AMA3707 Real Analysis

1 Sets and Functions

Exercise 1.1. [exc:abbott-1-2-3] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.3.

(a) If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ and every A_n is infinite, then $\bigcap_{n=1}^{\infty} A_n$ is infinite.
Given that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ are all containing infinite number of elements.
This is a nested sequence that

$$\bigcap_{n=1}^{m} A_n = A_m$$

Notice that for all sets A_n, it contains infinite number of elements. Meaning that $m \to \infty$, A_m will still contain infinite number of elements.
Also,

$$\bigcap_{n=1}^{\infty} A_n = \lim_{m\to\infty} \bigcap_{n=1}^{m} A_n = A_m$$

So,

$$\left| \bigcap_{n=1}^{\infty} A_n \right| = |A_m| = \infty$$

Statement (a) is true.

(b) If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ and every A_n is finite and nonempty, then $\bigcap_{n=1}^{\infty} A_n$ is finite and nonempty.
By the same argument,

$$\bigcap_{n=1}^{\infty} A_n = \lim_{m\to\infty} \bigcap_{n=1}^{m} A_n = A_m, \quad 0 < |A_m| < \infty$$

So,

$$\bigcap_{n=1}^{\infty} A_n = A_m \quad \text{is finite and nonempty.}$$

Statement (b) is true.

(c) $A \cap (B \cup C) = (A \cap B) \cup C$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Define

$$D = A \cap (B \cup C), \quad E = (A \cap B) \cup C$$

Denote the above 8 conditions as $P_1, P_2, \ldots, P_8$.
By using definition of definition intersection and definition union, we can see that

- $\forall x \in P_1 \implies x \in D$ and $x \in E$
- $\forall x \in P_2 \implies x \in D$ and $x \in E$
- $\forall x \in P_3 \implies x \in D$ and $x \in E$
- $\forall x \in P_4 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_5 \implies x \notin D$ and $x \in E$
- $\forall x \in P_6 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_7 \implies x \notin D$ and $x \in E$
- $\forall x \in P_8 \implies x \notin D$ and $x \notin E$

So, $D = P_1 \cup P_2 \cup P_3$ and $E = P_1 \cup P_2 \cup P_3 \cup P_5 \cup P_7$.
$\exists x, x \in E \land x \notin D$ such as $x \in P_5$ or $x \in P_7$, meaning that $D \subset E$.
According to the definition of set equality, $D \neq E$.
Statement (c) is false.

(d) $A \cap (B \cap C) = (A \cap B) \cap C$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Define

$$D = A \cap (B \cap C), \quad E = (A \cap B) \cap C$$

Denote the above 8 conditions as $P_1, P_2, \ldots, P_8$.
By using definition of definition intersection and definition union, we can see that

- $\forall x \in P_1 \implies x \in D$ and $x \in E$
- $\forall x \in P_2 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_3 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_4 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_5 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_6 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_7 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_8 \implies x \notin D$ and $x \notin E$

So, $\forall x \in D \implies x \in E$, ie $D \subseteq E$. Also, $\forall x \in E \implies x \in D$, ie $E \subseteq D$.
According to the definition of set equality, $D = E$.
Statement (d) is true.

(e) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Exercise 1.2. [exc:abbott-1-2-7] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.7.

Exercise 1.3. [exc:abbott-1-2-9] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.9.

2 Suprema and Infima

Exercise 2.1. [exc:abbott-1-3-3] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.3.

Exercise 2.2. [exc:abbott-1-3-4] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.4.

Exercise 2.3. [exc:abbott-1-3-11] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.11.